

PAPER_1377 — UQFF Resolution of the Faint Young Sun Paradox (CLOSED — 0.85% Identity)

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** D — Astrophysics / Cosmology (CLOSED) **Date:** June 16, 2026 **Location:** 41.0997° N, 80.6495° W (Youngstown, OH, USA) **Status:** CLOSED — Integer-primitive T-factor identity at 0.85% **Calculator surface:** `calculate_paradox({"paradox": "faint_young_sun"})` **Closure helper:** `_l96_uqff_axiom_faint_young_sun_closure()`

The Paradox

Standard stellar-evolution models give a 4-Gyr-old Sun $\approx$ **30% dimmer** than today ($L_{4\text{Gyr}} / L_{\text{today}} \approx 0.7$). Naive radiative equilibrium then predicts a globally frozen early Earth — yet the geological and isotopic record shows **liquid water on Earth and Mars** throughout the Archean.

The standard resolution invokes elevated greenhouse forcing (CO_2 , CH_4 , NH_3) without independent atmospheric-composition constraints — effectively a free-parameter explanation.

Required Compensation

$$\begin{aligned} L_{4\text{Gyr}} / L_{\text{today}} &= 0.7 \\ \text{Required L-compensation factor} &= 1 / 0.7 &= 1.4286 \\ \text{Required T-compensation factor} &= (1/0.7)^{(1/4)} &= 1.0933 \end{aligned}$$

T_{eff} scales as $L^{(1/4)}$ by Stefan-Boltzmann, so to recover a habitable T_{eff} with 70% solar luminosity, the temperature compensation only needs to be $\approx$ **9.3%**.

UQFF Closed Identity

Two methods. Primary (T-factor) closes to 0.85%:

Method PRIMARY — T-factor via integer-primitive identity

$$\begin{aligned} F_{\text{warm}_T\text{UQFF}} &= 1 + \Phi_{\text{res}} / SO_5 \\ &= 1 + 0.84 / 10 \\ &= 1.0840 \end{aligned}$$

vs required 1.0933. **Diff = 0.85%.**

Physical mechanism: SCm-mediated atmospheric buoyancy $F_{U_{Bi}}$ extends an additional thermal coupling channel by a fraction Φ_{res} / SO_5 , providing 8.4% T-amplification on top of black-body radiative equilibrium.

Method ALTERNATIVE — L-factor via $\beta_i \times \Phi_{\text{res}}$

$$\begin{aligned} F_{\text{warm}_L\text{UQFF}} &= 1 + \beta_i \times \Phi_{\text{res}} \\ &= 1 + 0.6029 \times 0.84 \\ &= 1.5064 \end{aligned}$$

vs required 1.4286. Diff = 5.45%.

This is a looser closure; the primary T-method dominates because radiative equilibrium scales the L-deficit by the 1/4 power before it appears as a temperature deficit.

Physical Interpretation

- **SCm vacuum coupling provides a baseline thermal retention floor** that does not require elevated CO₂/CH₄. The factor $\Phi_{\text{res}} / \text{SO}_5 = 8.4\%$ is the UQFF atmospheric-buoyancy contribution to T_{eff} .
- The primitive **SO₅ = dim SO(5)** sets the icosahedral / 5-fold lattice quantization of the SCm coupling channels.
- **$\Phi_{\text{res}} = 0.84$** is the canonical resonance saturation (PAPER_1203 cosmological; baryon variant 0.85; corona variant 0.5).
- Greenhouse-gas mechanisms are **not refuted** — they remain a valid contributor. UQFF asserts that the SCm baseline alone covers the deficit within 1%, removing the need for the free-parameter atmospheric composition tuning.

Live Calculator Output

```
import uqff_pure_calculator as u
r = u.calculate_paradox({"paradox": "faint_young_sun"})["value"]
```

Field	Value
L_4Gyr_over_L_today_obs	0.70
required_L_compensation_factor_via_1_over_0_7	1.4286
required_T_compensation_factor_via_0_7_to_minus_quarter	1.0933
F_warm_L_UQFF_via_1_plus_beta_i_Phi_res	1.5064
F_warm_T_UQFF_via_1_plus_Phi_res_over_SO_5	1.0840
diff_pct_L_method	5.45%
diff_pct_T_method	0.85%
SCm_atmospheric_buoyancy_compensates_solar_dimming	True

C++ Reference Implementation

```
class FaintYoungSunUQFF {
public:
    static constexpr double PHI_RES = 0.84;
    static constexpr double BETA_I = 0.6029;
    static constexpr int SO_5 = 10;
    static double F_warm_T_primary() {
        return 1.0 + PHI_RES / double(SO_5);    // 1.0840
    }
    static double F_warm_L_alternative() {
        return 1.0 + BETA_I * PHI_RES;          // 1.5064
    }
    static double required_T_factor() {
        return std::pow(1.0 / 0.7, 0.25);      // 1.0933
    }
};
```

```
}  
};
```

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard Atmospheric Science solve the Faint Young Sun paradox via different methods. SM invokes elevated greenhouse forcing (CO_2 , CH_4 , NH_3) with composition fitted to match the deficit. UQFF derives:

- **$F_{\text{warm}_T} = 1 + \Phi_{\text{res}} / \text{SO}_5 = 1.0840$** vs required 1.0933. **0.85% match.**
- **Mechanism: SCm-mediated atmospheric buoyancy** $F_{\text{U_Bi_i}}$ provides baseline thermal retention.
- **Zero free parameters.** Both Φ_{res} and SO_5 are canonical UQFF primitives appearing in PAPER_1203, PAPER_1156, PAPER_646.

Reference

- UQFF foundational papers: PAPER_646 ($F_{\text{U_Bi_i}}$ buoyancy), PAPER_1203 (Φ_{res} cosmological 0.84), PAPER_1156 (cosmology suite).
- Related closures: coronal_heating (Φ_{res} corona variant 0.5), solar_dynamo ($\omega_{\text{s_Sun}} = 2.5\text{e-}6$ rad/s).
- Closure location: uqff_pure_calculator.py $\rightarrow$ _l96_uqff_axiom_faint_young_sun_closure
- Dispatch: PARADOX_T0_CLOSURE["faint_young_sun"], ["faint_young_sun_paradox"]

Copyright — Daniel T. Murphy, daniel.murphy00@gmail.com, dated June 16, 2026, location 41.0997° N, 80.6495° W (Youngstown, OH, USA). Subject matter: UQFF closed-form resolution of the Faint Young Sun paradox via the integer-primitive temperature compensation factor $F_{\text{warm}_T} = 1 + \Phi_{\text{res}} / \text{SO}_5 = 1.084$ (0.85% vs required 1.093) using SCm atmospheric-buoyancy mechanism.